

VAISHNAVI ATHREY RAMESH

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PROFESSIONAL SUMMARY

Software engineer with 2.5+ years of experience in backend development, distributed systems, and cloud-native architectures. Proficient in Java, Go, Python, and C++; experienced with Spring Boot, FastAPI, and Docker for scalable backend services. Hands-on expertise with AWS (Redshift, Lake Formation, CDK, IAM) and PostgreSQL/MongoDB-based data systems, combined with growing experience in RAG pipelines, CLIP-based multimodal retrieval, and agentic AI tooling through independent and coursework projects.

SKILLS

Languages: Java, Python, C/C++, Go, SQL, JavaScript, HTML, CSS

Frameworks & Tools: Spring Boot, FastAPI, Django, Angular, Next.js, LangChain, Docker, Kubernetes, Selenium

Libraries: Pandas, NumPy, Matplotlib, TensorFlow, Keras, PyTorch, Scikit-Learn, CLIP, LoRA/PEFT

Databases & Cloud: PostgreSQL, MySQL, MongoDB, Redis, Cassandra, Neo4j, AWS (Redshift, S3, IAM, KMS, Lake Formation, Athena, CDK)

EXPERIENCE

Amazon — Software Developer Summer Intern

06/2026 – 08/2026

- Built a Redshift/Athena-backed data-integrity monitoring tool using two cross-service entropy checks for the accounting pipeline, validating 8M+ production records with zero false positives.
- Replaced a legacy bulk SQL JOIN with a GraphQL-based lookup in a Java service-layer feature handling 5,000+ transactions/sec, and provisioned the supporting AWS infrastructure via CDK (Redshift Serverless, Lake Formation, S3, IAM, KMS).
- Designed a General Ledger AWS pipeline integrating a previously uncaptured financial source into the UFL ledger, analyzing 1,700+ accounting source types across 87B+ records to close reconciliation gaps against the company-wide P&L.

IBM — Software Developer I

04/2024 – 07/2025

- Migrated the persistence layer of distributed Go and Java microservices (Guardium Insights Risk Analytics Engine) from MongoDB and DB2 to PostgreSQL and AWS Redshift, redesigning normalized schemas and converting GORM/JDBC data-access code to improve query performance by 25% and cut storage costs by 34%.
- Built a DB2/PostgreSQL dual-database compatibility layer for the Risk Analytics Engine, replacing DB2-specific JDBC calls and rewriting dynamically generated SQL to keep queries portable across both engines in production.
- Root-caused cross-service failures in a multi-tenant risk-analytics pipeline spanning Kafka event delivery and gRPC/protobuf APIs, tracing a DB2 column-resolution error through dynamically generated tenant-scoped SQL to its source in the query generator.

IBM — Software Developer Intern

01/2023 – 07/2023

- Developed a responsive Angular-based account creation and onboarding flow for a digital banking product, reducing user drop-off by 20%.
- Built Java/Spring Boot REST APIs backed by MySQL to power core banking transactions, adding request validation and service-layer logic for user account and payment workflows.
- Took ownership of the remaining project scope after a teammate's departure left the 3-person intern team at two, splitting the workload with the remaining intern and leading delivery of the onboarding flow and REST APIs on schedule.

PROJECTS

EEG-Based Object Classification & CLIP Retrieval – CMU Deep Learning Course Project

04/2026

Python, PyTorch, CNN + Transformer, CLIP (ViT-B/32), LoRA/PEFT, SLURM HPC Cluster

- Built a CNN+Transformer-based EEG object-classification pipeline on a 13-subject, 122-channel dataset (20 PASCAL VOC categories), alongside a CLIP ViT-B/32-backed image-to-caption retrieval baseline (R@1 27.61%) to benchmark EEG-based retrieval against.
- Root-caused a silent architecture mismatch in a CLIP-alignment fine-tuning pipeline via systematic state-dict diffing (77 copied, 26 shape-mismatched, 22 name-mismatched tensors), tracing it to two incompatible CNN architectures blocking warm-start transfer.
- Diagnosed a model-collapse failure in a four-loss CLIP-alignment training run through embedding-geometry analysis (pairwise cosine similarity), then resolved it by reweighting the loss terms.

DocuMind AI

12/2025 – 02/2026

Python, Docker, FastAPI, LangChain, RAG, Pinecone, Next.js, Tailwind CSS

- Engineered an end-to-end Retrieval-Augmented Generation (RAG) pipeline using pdfplumber-based PDF ingestion and structured clause extraction to automate legal-document review.
- Implemented OpenAI-embedding-backed semantic search with Pinecone vector storage and LangChain-orchestrated retrieval (512-token chunks, 50-token overlap), achieving 92% retrieval accuracy on benchmark queries.

EDUCATION

Carnegie Mellon University

Pittsburgh, PA | 08/2025 – 12/2026

Master of Information Systems Management

Coursework: Database Management Systems, Distributed Systems, Deep Learning, NoSQL, Operationalizing AI, Machine Learning, Large Language Model Systems

R.V. College of Engineering

Bengaluru, India | 08/2019 – 08/2023

B.E., Electronics and Instrumentation Engineering

Coursework: Data Structures, C/C++, Operating Systems, Python, Data Networks, Wireless Technologies, Internet of Things